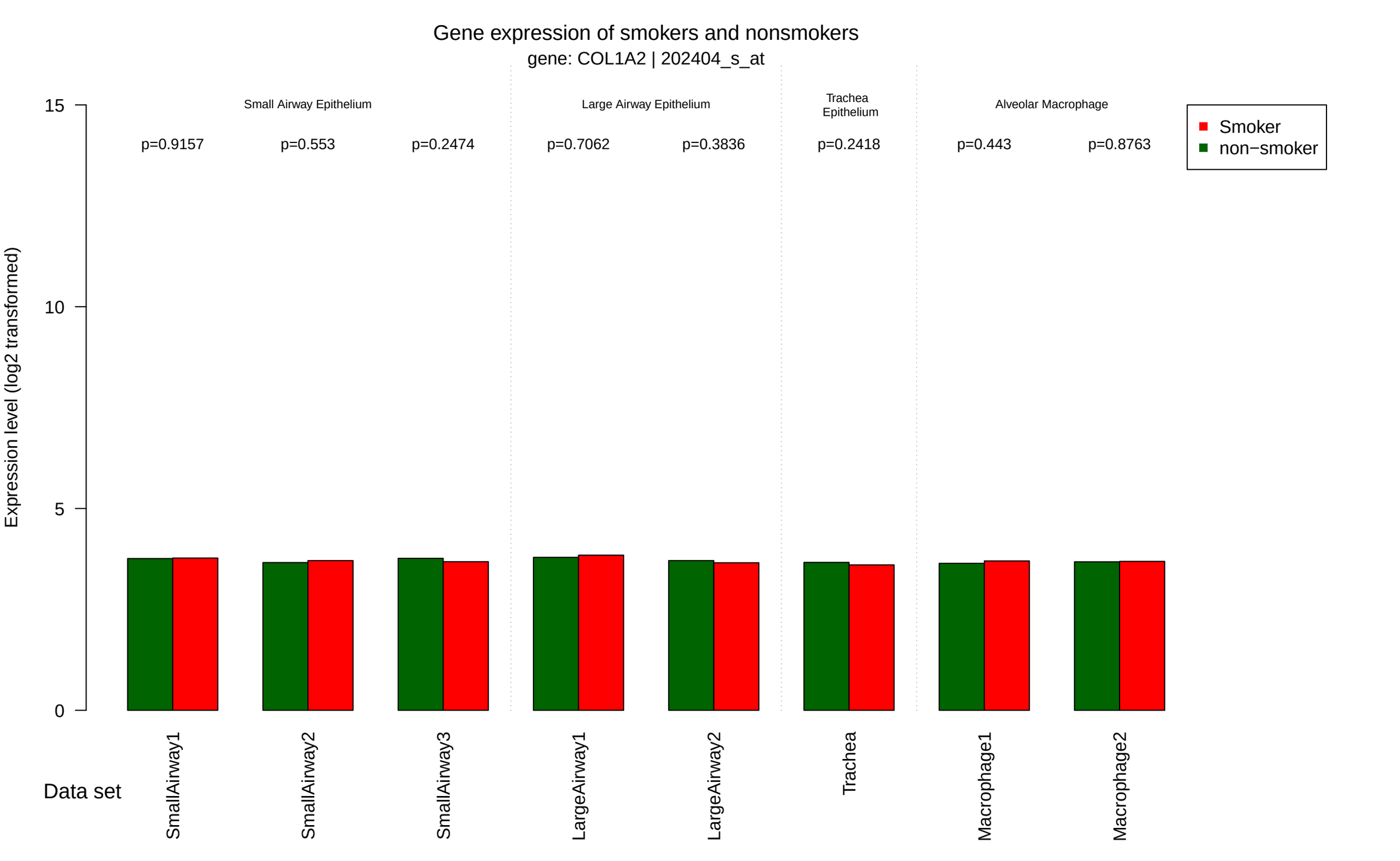
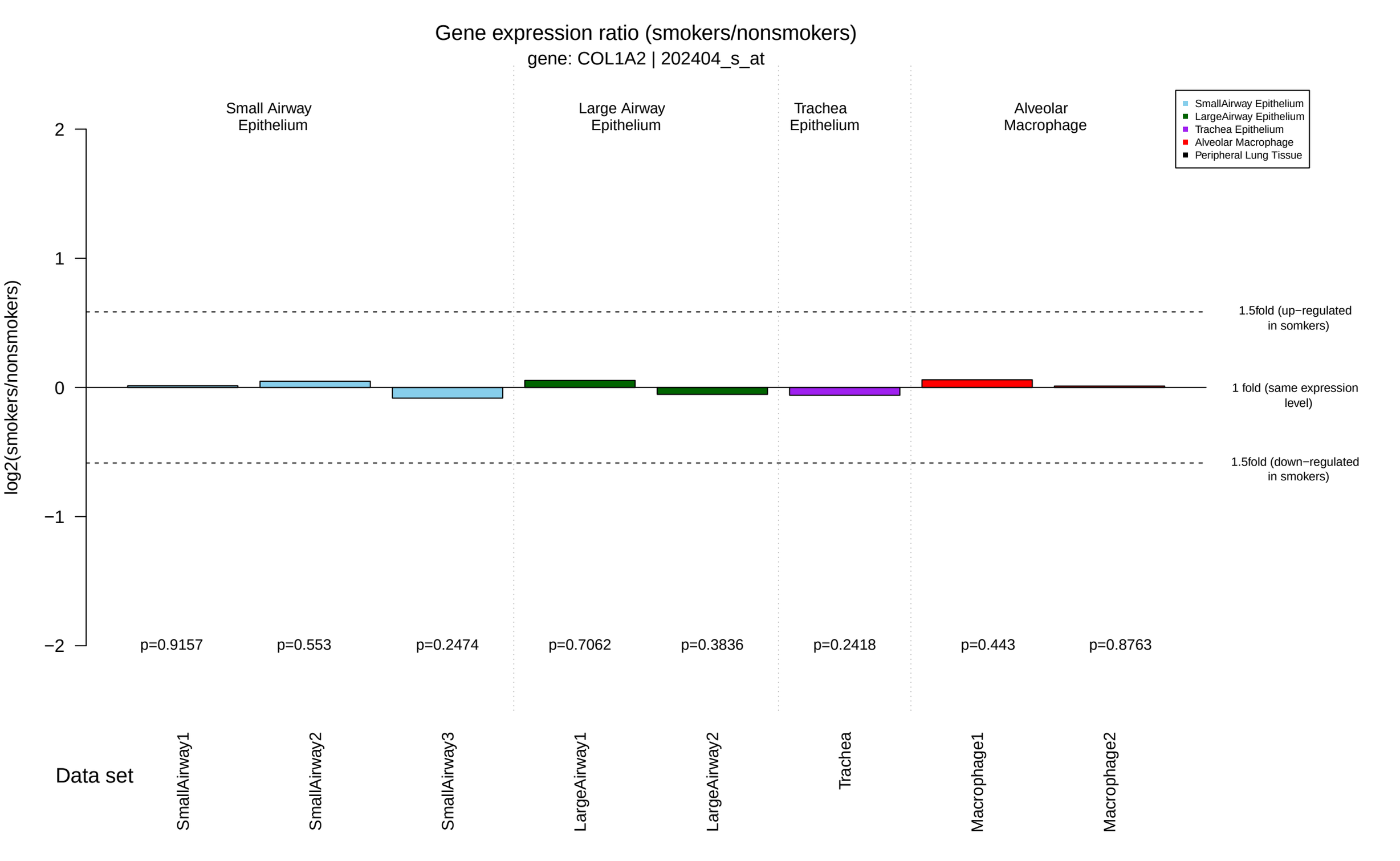
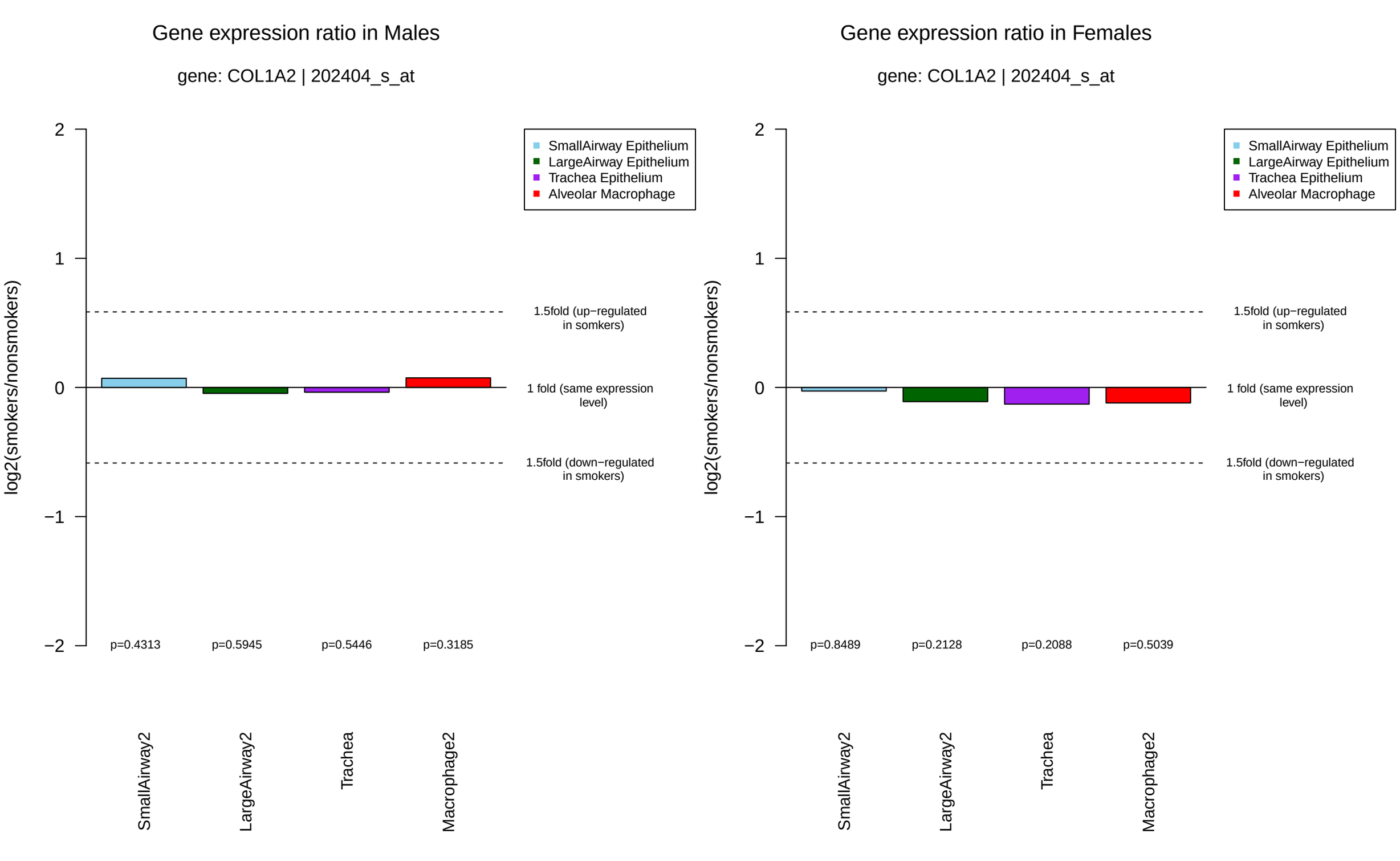
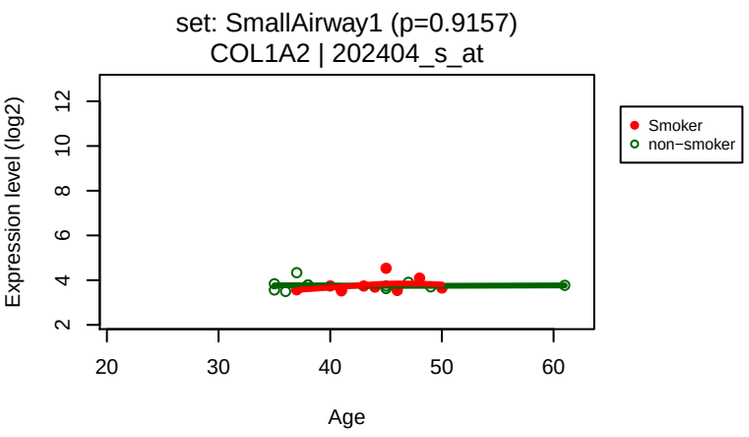
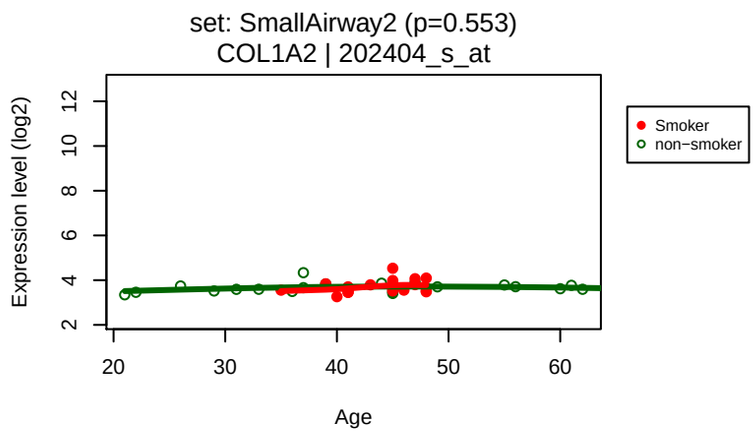
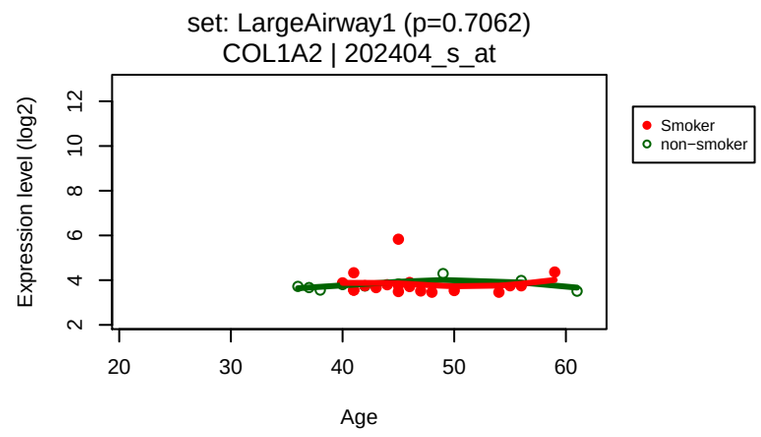

Gene expression of smokers and nonsmokers
gene: COL1A2 | 202404_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9157
p=0.553
p=0.2474
p=0.7062
p=0.3836
p=0.2418
p=0.443
p=0.8763
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

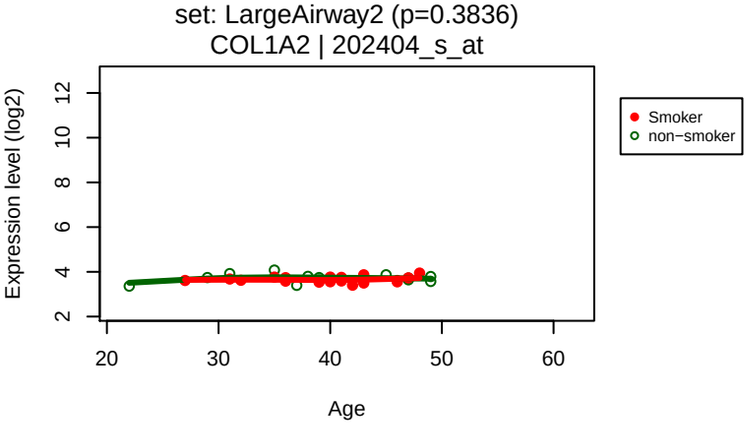

Gene expression ratio (smokers/nonsmokers)
gene: COL1A2 | 202404_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9157
p=0.553
p=0.2474
p=0.7062
p=0.3836
p=0.2418
p=0.443
p=0.8763
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

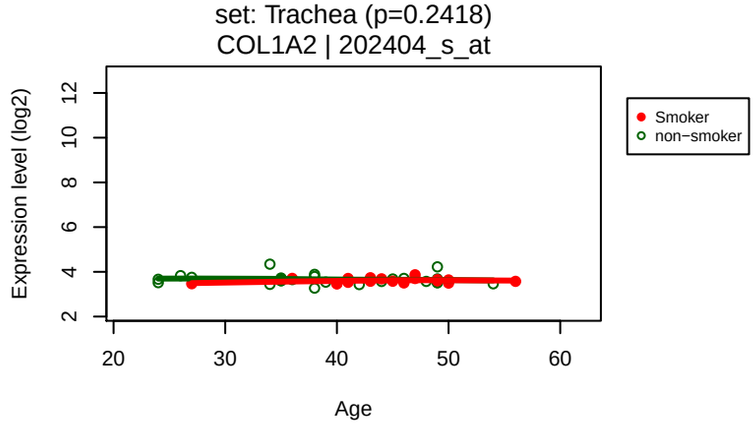

Gene expression ratio in Males
gene: COL1A2 | 202404_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4313
p=0.5945
p=0.5446
p=0.3185
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL1A2 | 202404_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8489
p=0.2128
p=0.2088
p=0.5039
SmallAirway2
LargeAirway2
Trachea
Macrophage2

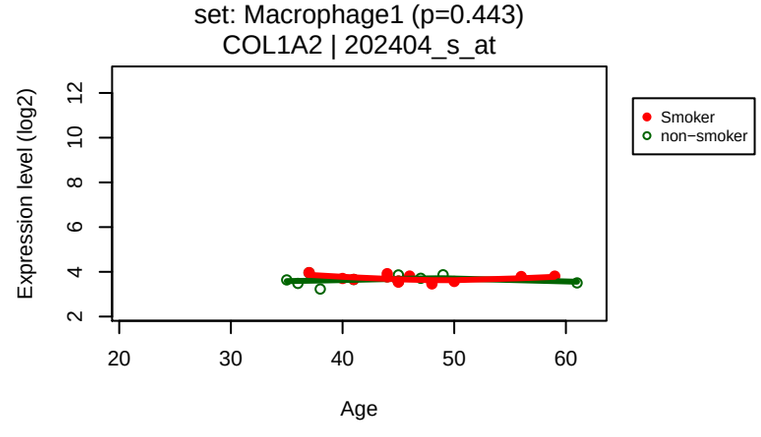

set: SmallAirway1 (p=0.9157)
COL1A2 | 202404_s_at
Expression level (log2)
Age
Smoker
non−smoker

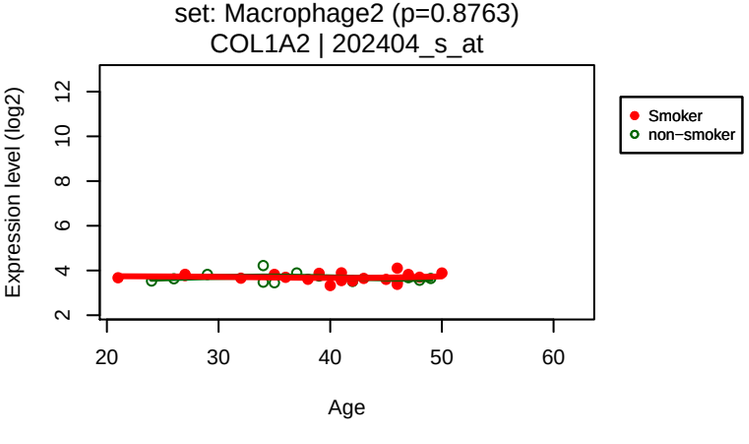

set: SmallAirway2 (p=0.553)
COL1A2 | 202404_s_at
Expression level (log2)
Age
Smoker
non−smoker

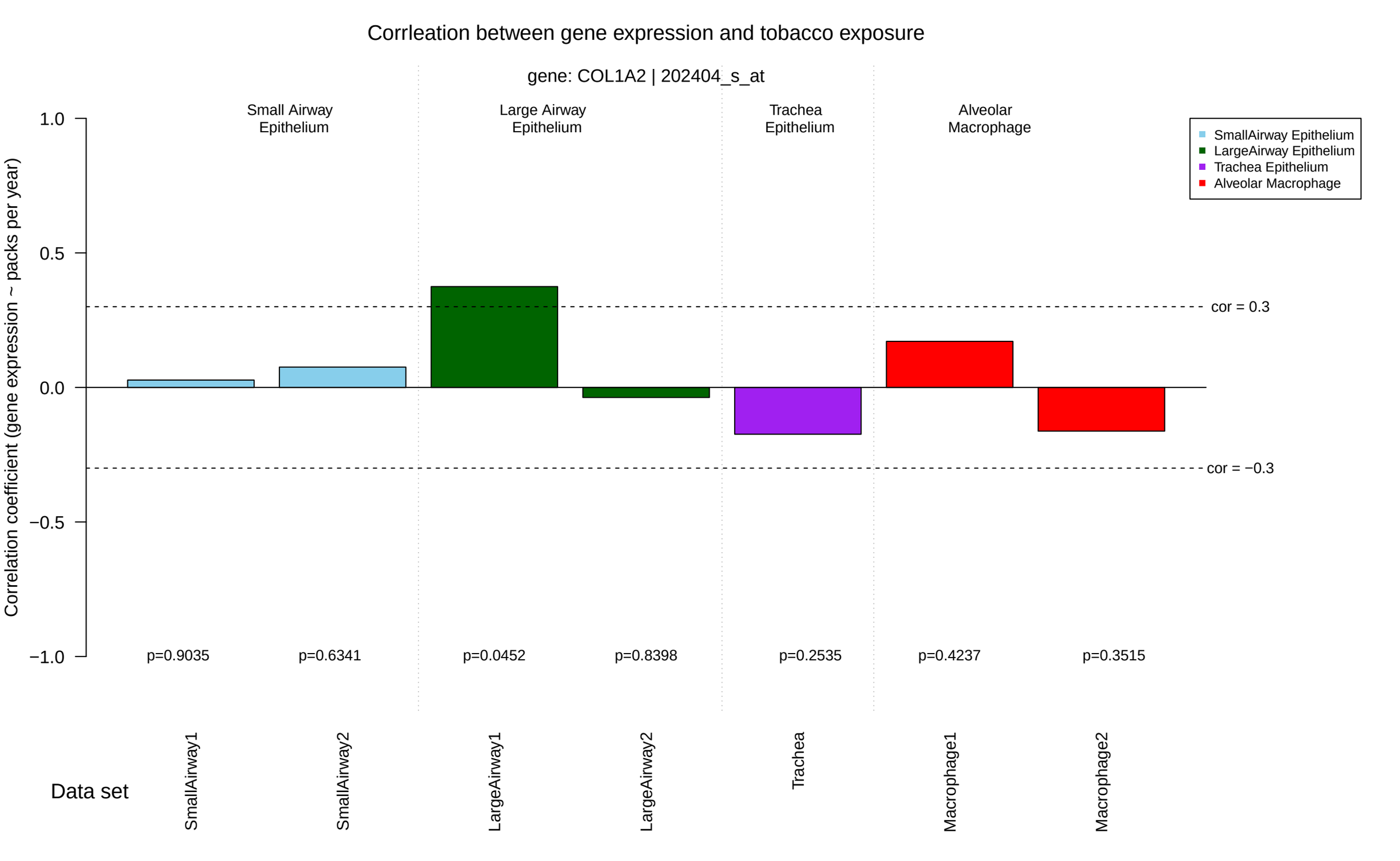

set: LargeAirway1 (p=0.7062)
COL1A2 | 202404_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3836)
COL1A2 | 202404_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2418)
COL1A2 | 202404_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.443)
COL1A2 | 202404_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8763)
COL1A2 | 202404_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL1A2 | 202404_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9035
p=0.6341
p=0.0452
p=0.8398
p=0.2535
p=0.4237
p=0.3515
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2